

CHL5233: Quarto in-class assignment

Countries by region and sub-region

Daniel Guo

Read in data and create an analytical data set

The supplied files describe the study country list and geographic classifications. The files are stored in `data/raw/` in this project; paths below are relative to this report in `reports/`.

```
countries <- read.table("../data/raw/countries.txt", header = TRUE)
dim(countries)
```

```
[1] 186    2
```

The countries data set contains **186 countries** and **2 columns**. The supplied country list contains 186 distinct ISO codes. This report uses all countries in the supplied file.

```
regions <- read.table("../data/raw/regions.txt", header = TRUE)
dim(regions)
```

```
[1] 249    6
```

The regions data set contains **249 country or territory entries** and **6 columns**.

We match the three-letter ISO codes, avoiding differences in country names. `semi_join()` retains only entries in `regions` whose codes occur in `countries`.

```
study_regions <- regions |>
  semi_join(countries, by = c("country_code" = "ISO"))
nrow(study_regions)
```

```
[1] 186
```

All **186 countries** matched a geographic entry; none were lost or duplicated in the match.

Tabulate countries by region and sub-region

```
region_counts <- study_regions |>
  group_by(region) |>
  summarise(count = n(), .groups = "drop") |>
  arrange(desc(count), region)

kable(region_counts, col.names = c("Region", "Count"),
      caption = "Study countries by region", booktabs = TRUE, label = NA) |>
  kable_styling(latex_options = "HOLD_position")
```

Table 1: Study countries by region

Region	Count
Africa	54
Asia	46
Europe	40
Americas	35
Oceania	11

Countries by sub-region

```
subregion_counts <- study_regions |>
  group_by(region, sub_region) |>
  summarise(count = n(), .groups = "drop") |>
  arrange(desc(count), sub_region)

kable(select(subregion_counts, sub_region, count),
      col.names = c("Sub-region", "Count"),
      caption = "Study countries by sub-region", booktabs = TRUE, label = NA) |>
  kable_styling(latex_options = "HOLD_position")
```

Table 2: Study countries by sub-region

Sub-region	Count
Sub-Saharan Africa	48
Latin America and the Caribbean	33
Western Asia	16
Southern Europe	13
South-eastern Asia	11
Eastern Europe	10
Northern Europe	10
Southern Asia	9
Western Europe	7
Northern Africa	6
Central Asia	5
Eastern Asia	5
Melanesia	4
Micronesia	3
Australia and New Zealand	2
Northern America	2
Polynesia	2

Both tables total **186 countries**. Africa has the largest regional count (54), and Sub-Saharan Africa has the largest sub-regional count (48).

Data visualization using ggplot

The regional chart summarizes the overall distribution. Each bar counts individual countries in the analytical data set.

```
ggplot(study_regions,
       aes(x = fct_infreq(region))) +
  geom_bar(fill = "#287D8E", width = 0.7) +
  labs(x = "Region", y = "Number of countries") +
  scale_y_continuous(breaks = seq(0, 60, 10)) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank())
```

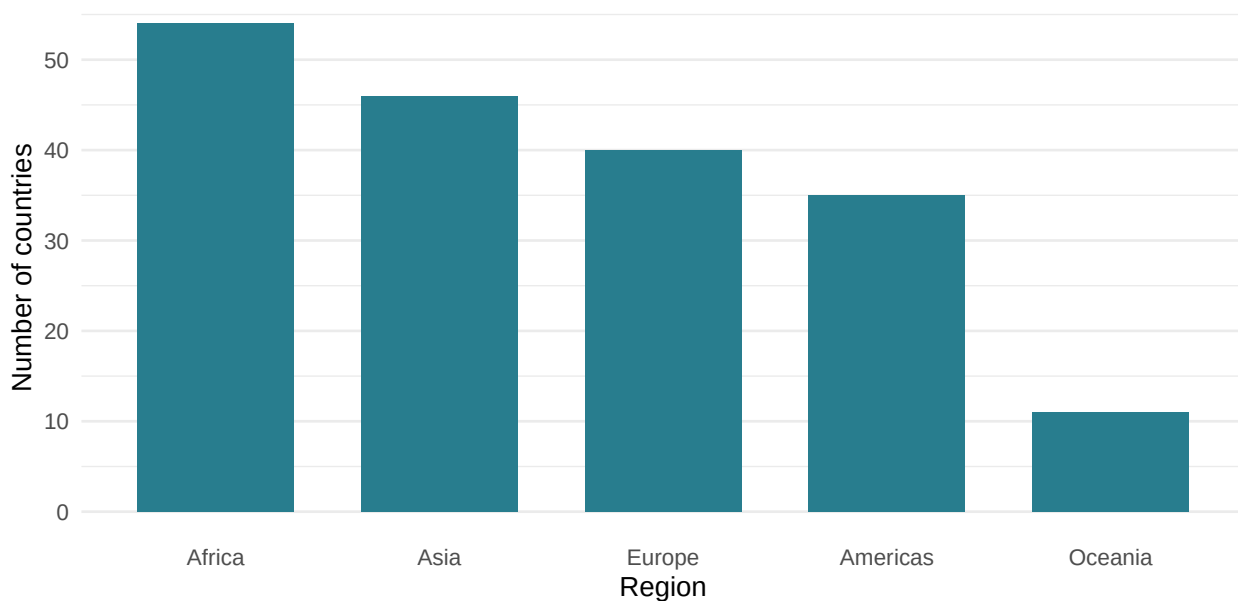


Figure 1: Number of study countries in each region.

The sub-regional chart uses horizontal bars, colored by region. Reversing the frequency ordering places the largest count at the top.

```
ggplot(study_regions,
       aes(x = fct_rev(fct_infreq(sub_region)), fill = region)) +
  geom_bar(width = 0.75) +
  coord_flip() +
  scale_fill_manual(values = c(
    Africa = "#0072B2", Americas = "#D55E00", Asia = "#009E73",
    Europe = "#CC79A7", Oceania = "#E69F00"
  )) +
  scale_y_continuous(breaks = seq(0, 50, 10)) +
  labs(x = "Sub-region", y = "Number of countries", fill = "Region") +
  theme_minimal(base_size = 10) +
  theme(legend.position = "bottom",
        panel.grid.major.y = element_blank())
```

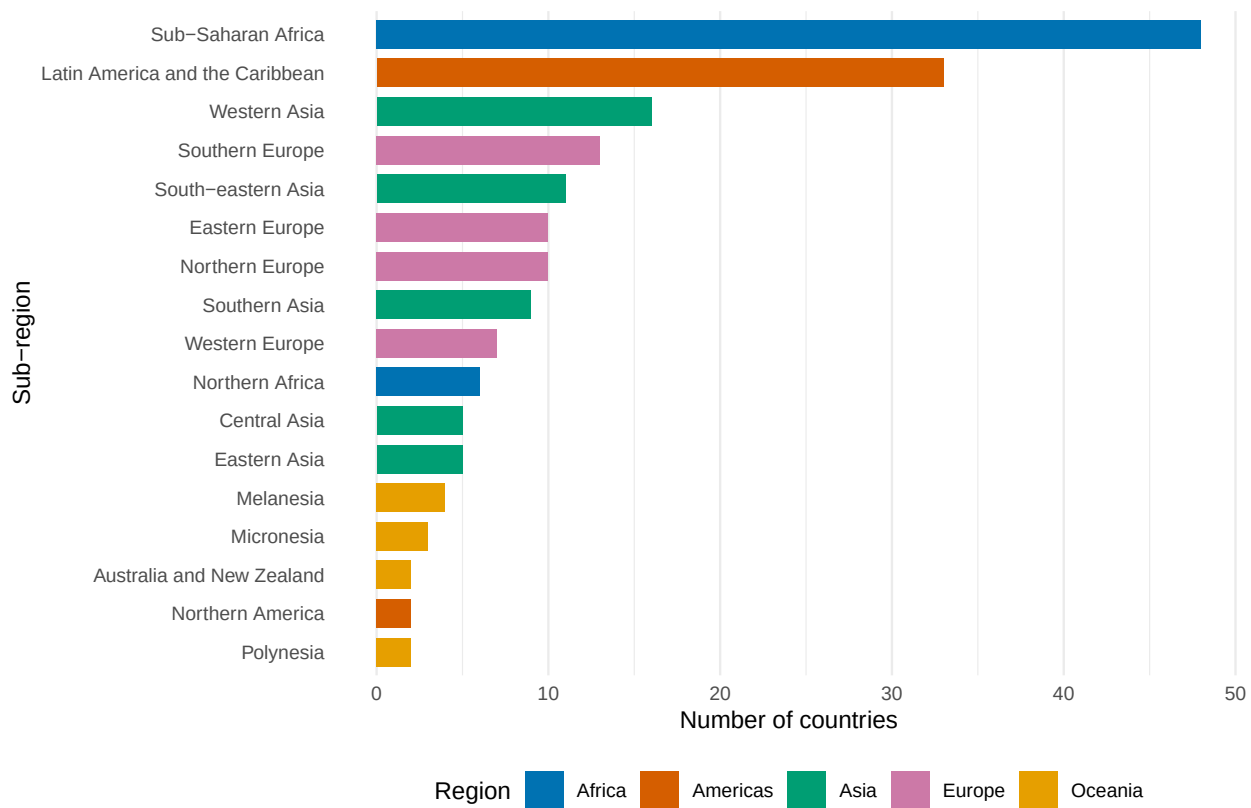


Figure 2: Study countries by sub-region, ordered by count and colored by region.

These counts describe geographic representation in the supplied list, rather than the number or severity of armed conflicts.